

Bandit Telephone Service Customer Churn Analysis

Business Understanding

In our analytical report on the Bandit Telephone Service company's customer churn, we set out to answer the question, "Which service and account features are most associated with a customer's decision to leave?" The objective of this report will be to identify the factors that are most associated with customer churn. Identifying which features are key in contributing to customer loss will help us to figure out who the high-risk customers are, and subsequently will allow us to develop retention strategies for reducing customer churn. We will also be applying the insights from our analysis to provide suggestions for the development of targeted retention strategies.

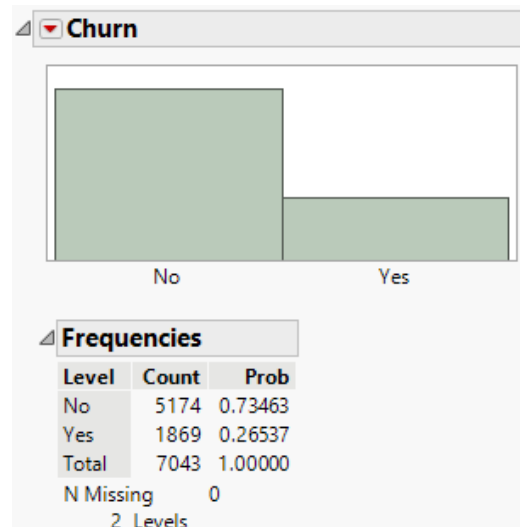
We will be using a dataset containing data on many service and account features associated with the company's customers. We will be using decision tree modeling with the JMP software in order to identify the variables that are significantly associated with customer churn. The goal of our decision tree modeling will be to classify our data into a hierarchical structure of rules based on the customer churn data, allowing us to pinpoint the features that are more associated with customer churn. The decision tree model will visualize for us the variables that are not significant in identifying high-risk customers and the ones that are, and therefore is appropriate for figuring out which of the features are key in potentially impacting churn.

Data Understanding

The Bandit Telephone Service Customer Churn dataset contains 7,043 data points, with each data point representing a current or past customer. There are 21 fields or variables, including customer ID and whether or not the customer churned, with each of the variables describing a different service or account feature related to the customer. The dataset was received as a CSV file, but was converted to a JMP file format to utilize in the

JMP software. For our report, we will be examining the customer churn data, represented by the variable Churn in our dataset, and its relation to what we have determined to be potentially key variables associated with customer churn. We will explain the standards by which we selected these potentially key variables shortly.

Figure 1 – Churn Distribution and Statistics



In this figure, we see the distribution of the churn data in our dataset, along with a few important statistics. The churn data is a binary variable split into either “Yes” for the customer having churned or “No” for the customer not having churned. We note that roughly 73.5% of the customers have not churned, while a minority, roughly 26.5%, have churned. Although the majority have not churned, 26.5% would be considered a significant proportion of any company’s customer base to lose. There are no missing values in this dataset, showing that we possess churn data for every customer in our dataset.

Initial Observations – Selecting Potential Predictor Variables

Redundancy

Some of the variables were redundant when compared with our chosen key variables, and therefore we decided that they were not necessary for our model. For example,

TotalCharges (total amount billed to the customer) was deemed redundant with one of our key variables, MonthlyCharges (monthly charges paid by the customer).

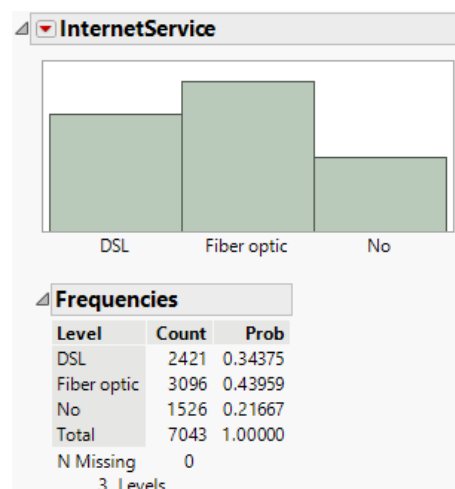
Domain Knowledge

Some of the variables were considered relevant due to our knowledge of what are currently considered significant features in the telecommunications market, such as PaperlessBilling (whether the customer has paperless billing or not), and thus were deemed relevant to our analysis.

Potentially Significant Variables

A select few of the variables were considered to be potentially significant for predicting customer churn: InternetService (type of internet service), OnlineSecurity (whether the customer has online security or not), Contract (type of contract), PaperlessBilling (whether the customer has paperless billing or not), and MonthlyCharges (monthly charges paid by the customer). These variables passed our initial filtering requirements, and were deemed relevant to customer churn based on our domain knowledge regarding the telecommunications market.

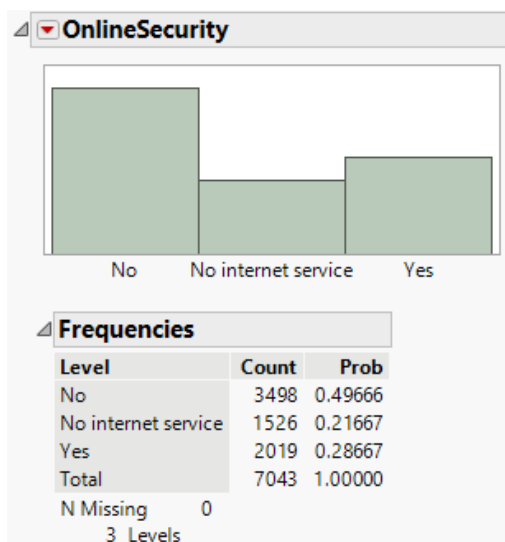
Figure 2 – Internet Service Distribution and Statistics



In this figure, we see the distribution of the internet service data in our dataset, along with a few important statistics. The InternetService variable is a categorical variable split into

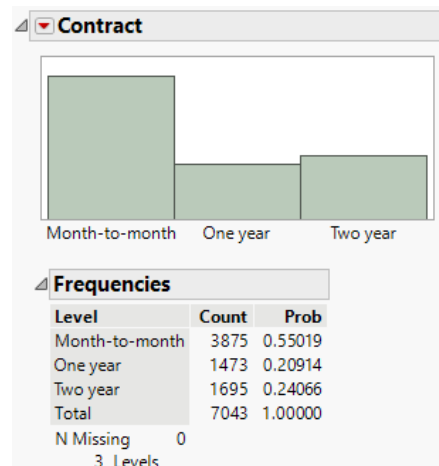
DSL, Fiber Optic, and No (representing no internet service). We note that roughly 34.4% of the customers have DSL, 44.0% have Fiber Optic, and 21.7% have no internet service. There are no missing values listed. We expect that one of the two types of internet service may potentially be associated with a lower churn rate due to providing better internet service overall, while we expect not having internet service to be associated with relatively higher churn rates since it shows that the customer is receiving less from and thus is likely less committed to the company's services.

Figure 3 – Online Security Distribution and Statistics



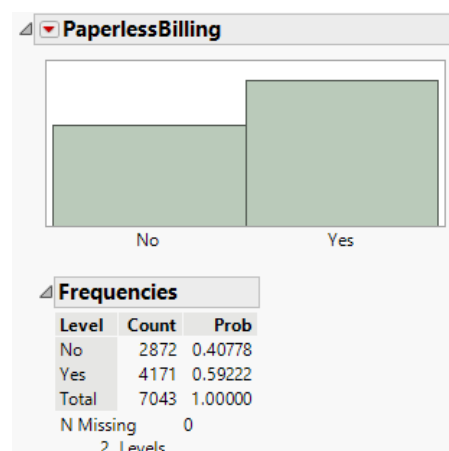
In this figure, we see the distribution of the online security data in our dataset, along with a few important statistics. The OnlineSecurity variable is a categorical variable split into Yes, No, and No internet service, with each value representing whether the customer has online security or not. We note that roughly 49.7% of the customers do not have online security, 21.7% have no internet service (and thus do not have online security either), and 28.7% have online security. There are no missing values listed. We expect that customers without online security will be associated with a higher rate of churn while those with online security will be associated with a lower rate of churn, because possessing the online security service indicates that the customer is receiving more from and thus is likely more committed to the company's services.

Figure 4 – Contract Distribution and Statistics



In this figure, we see the distribution of the contract data in our dataset, along with a few important statistics. The Contract variable is a categorical variable split into Month-to-month, One year, and Two year, with each value representing the type of contract based on duration. We note that roughly 55.0% of the customers have a month-to-month contract, 20.9% have a one-year contract, and 24.1% have a two-year contract. There are no missing values listed. We expect that customers with longer contracts will be associated with lower rates of churn, since the longer duration of the contract indicates that the customer is likely more committed to paying for and receiving the company's services.

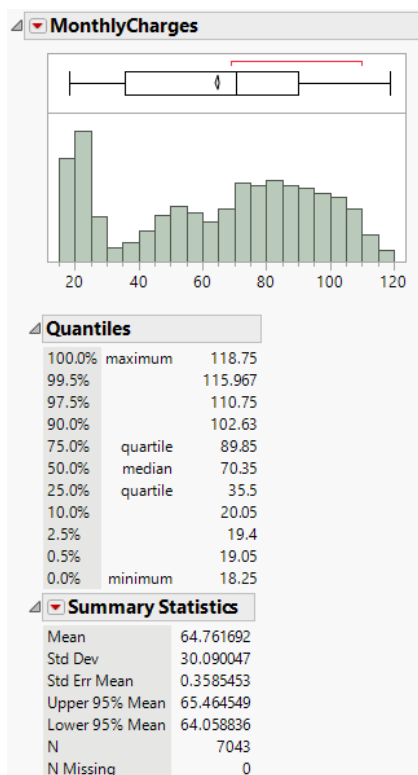
Figure 5 – Paperless Billing Distribution and Statistics



In this figure, we see the distribution of the paperless billing data in our dataset, along with a few important statistics. The PaperlessBilling variable is a binary variable split into Yes or

No, based on whether the customer has paperless billing or not. We note that roughly 40.8% of the customers do not have paperless billing, while roughly 59.2% of the customers do. There are no missing values listed. We expect that customers with paperless billing will be associated with lower rates of churn, since there has been more of a transition to paperless billing across commercial fields due to the relative efficiency that it provides as a means of billing.

Figure 6 – Monthly Charges Distribution and Statistics



In this figure, we see the distribution of the monthly charges data in our dataset, along with a few important statistics. The MonthlyCharges variable is a continuous variable with values representing the monthly charges paid by the customer. The monthly charges range from \$18.25 to \$118.75. The mean monthly charge is roughly \$64.76 with a standard deviation of roughly \$30.09, and a median of \$70.35. We observe that the distribution has a relatively high concentration at the lowest values, and yet possesses a significant proportion in the higher values as well. There are no missing values listed. We expect that higher monthly charges may be associated with high churn rate, due to customers

potentially wishing to switch to a different company's services where they would perhaps be charged less.

Data Preparation

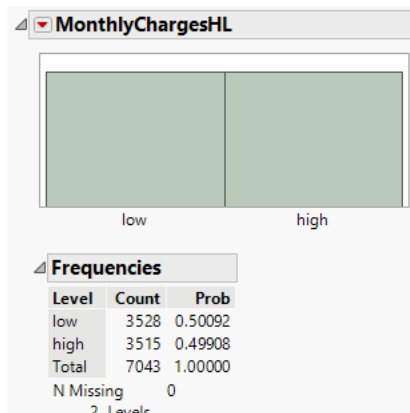
Our data preparation involved four steps. The first step was to confirm that Churn, the target variable, is binary. Upon reviewing Figure 1, we were able to verify that Churn is indeed a binary variable, split into Yes or No.

The second step was to convert the monthly charges variable into a categorical variable. We used the JMP software to develop a formula that converted a value of the monthly charges data to "high" if the value was greater than the median of the monthly charges data (which we identified as 70.35 from Figure 6), and to "low" if the value was at or lower than the median of the monthly charges data. This new binary categorical variable was labeled "MonthlyChargesHL".

The third step was to create a dummy variable for all of our variables. We used the JMP software to develop a formula that converted "No" to 0 and "Yes" to 1 for Churn and PaperlessBilling, and we labeled the dummy variables "ChurnD" and "PaperlessBillingD", respectively. For InternetService, we converted "No" to 0, "DSL" to 1, and "Fiber Optic" to 2 in a dummy variable labeled "InternetServiceD". For OnlineSecurity, "No" and "No internet service" were both converted to 0 since we are only testing for whether the customer has online security or not with this variable, and "Yes" was converted to 1 in a dummy variable labeled "OnlineSecurityD". For Contract, "Month-to-month" was converted to 0, "One year" was converted to 1, and "Two year" was converted to 2 in a dummy variable labeled "ContractD". Lastly, for MonthlyChargesHL, "low" was converted to 0 and "high" was converted to 1 in a dummy variable labeled "MonthlyChargesHLD".

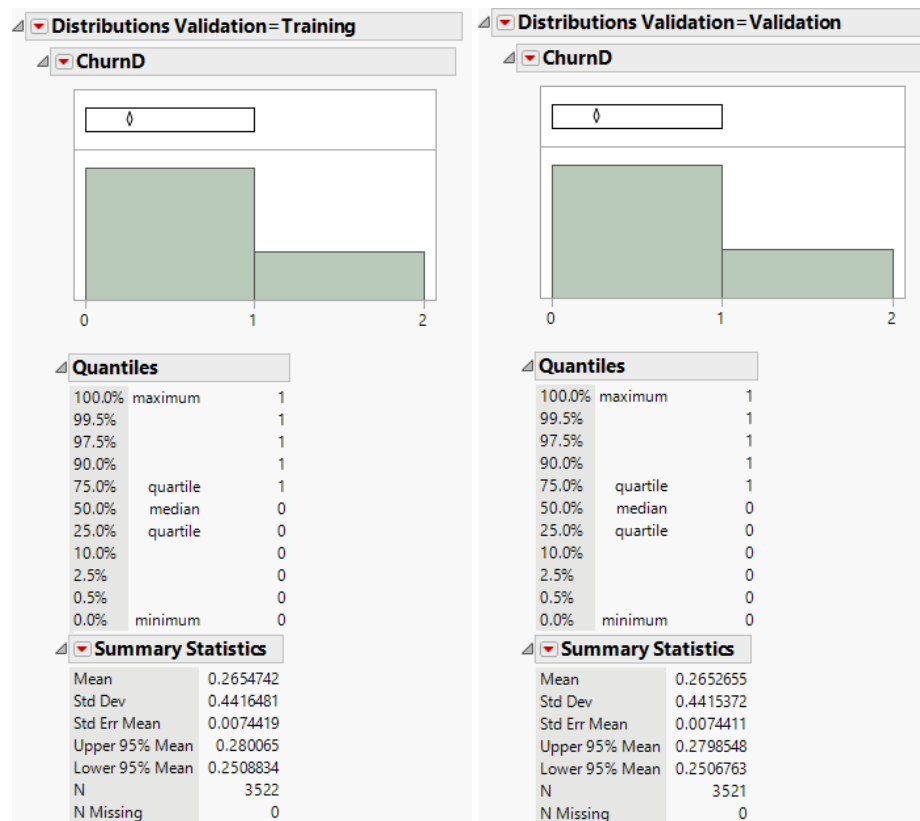
The fourth step was to split our data into a 50/50 training-validation split, using a random seed of 12345 for reproducibility. Afterwards, we examined both the training set and the validation set to confirm that the key variables had similar distributions in both sets.

Figure 7 – Converted Monthly Charges Distribution and Statistics



This figure shows the distribution of the monthly charges data after it was converted into a categorical variable. We observe that although the “high” and “low” values are roughly equal in count, there are a few more “low” values due to the “low” value including the median monthly charges values.

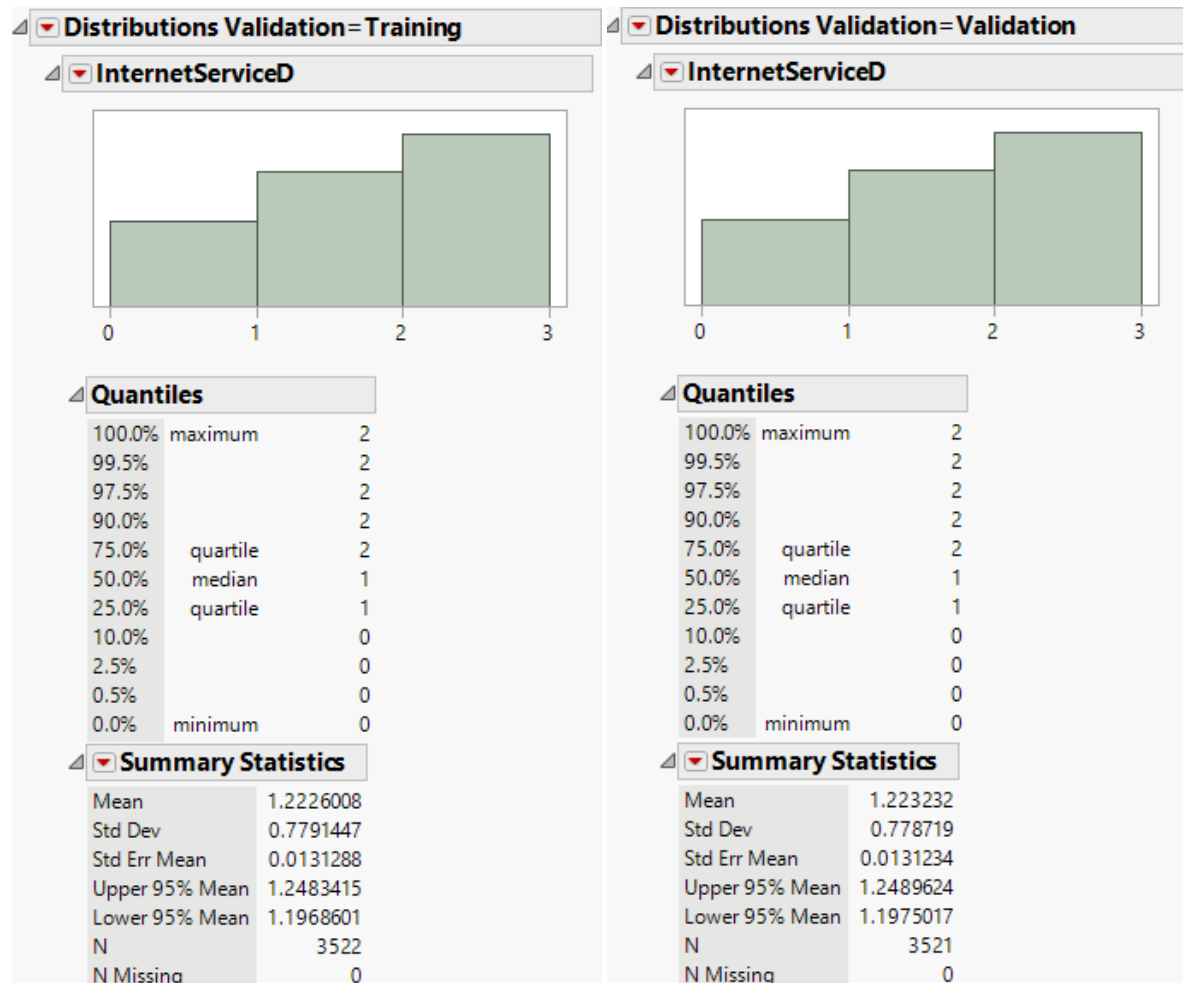
Figure 8 – Churn Dummy Variable Training-Validation Split



In this figure, we see the distributions of the dummy variable for Churn in both the training and validation sets. We see here that the distributions are virtually identical, with nearly the

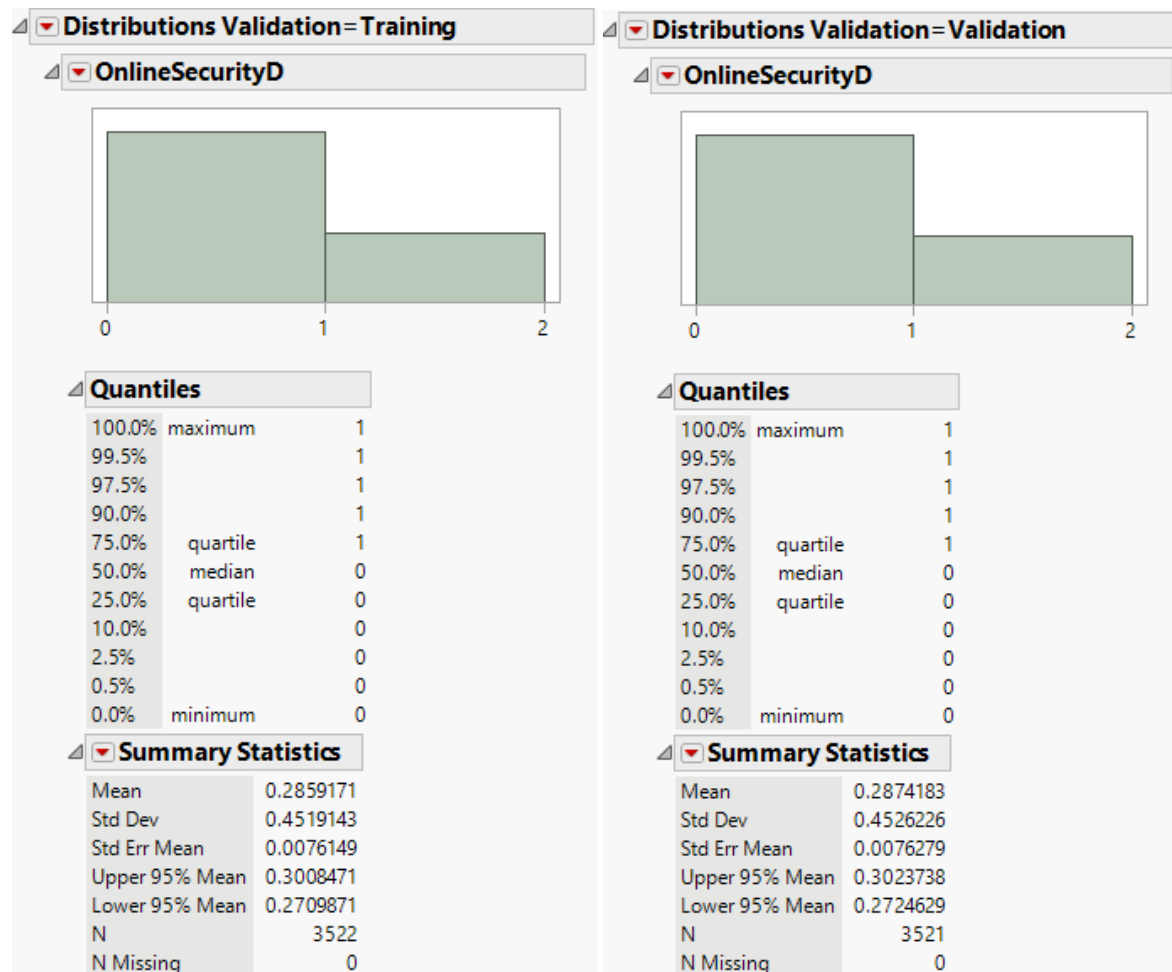
same counts (3522 in the training set and 3521 in the validation set) and nearly identical key statistics. We converted the dummy variable to a nominal type for our analysis.

Figure 9 – Internet Service Dummy Variable Training-Validation Split



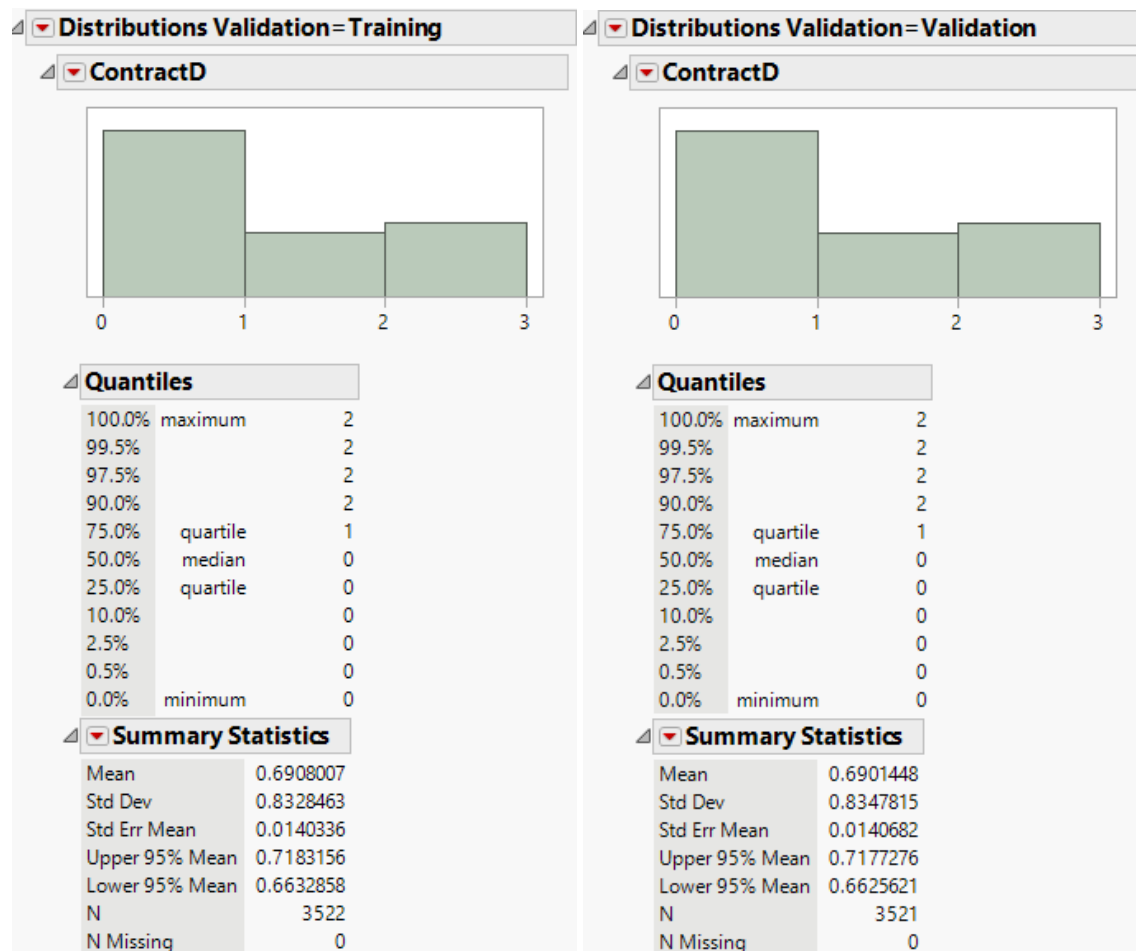
In this figure, we see the distributions of the dummy variable for Internet Service in both the training and validation sets. We see here that the distributions are virtually identical, with nearly the same counts (3522 in the training set and 3521 in the validation set) and nearly identical key statistics.

Figure 10 – Online Security Dummy Variable Training-Validation Split



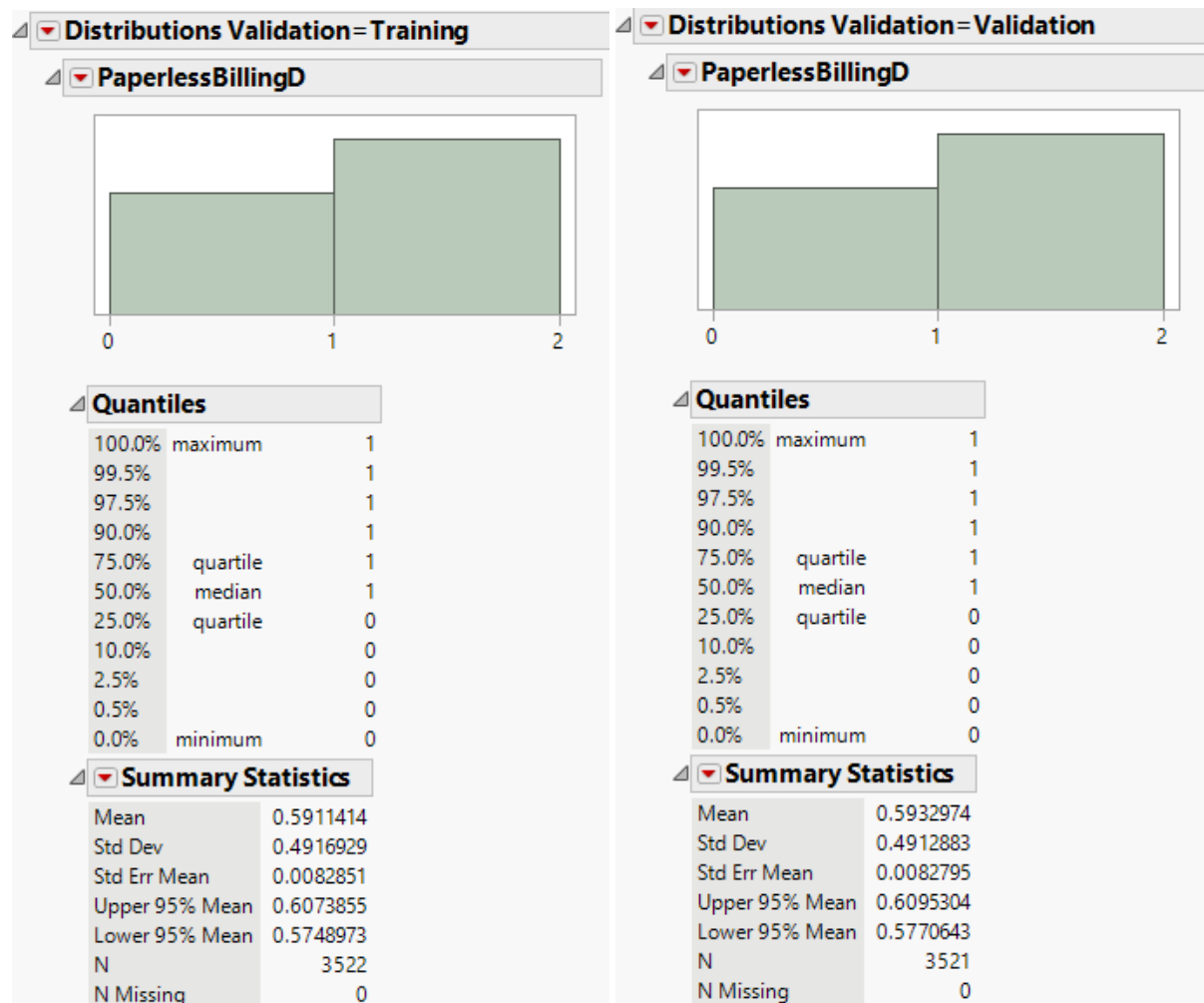
In this figure, we see the distributions of the dummy variable for Online Security in both the training and validation sets. We see here that the distributions are virtually identical, with nearly the same counts (3522 in the training set and 3521 in the validation set) and nearly identical key statistics.

Figure 11 – Contract Dummy Variable Training-Validation Split



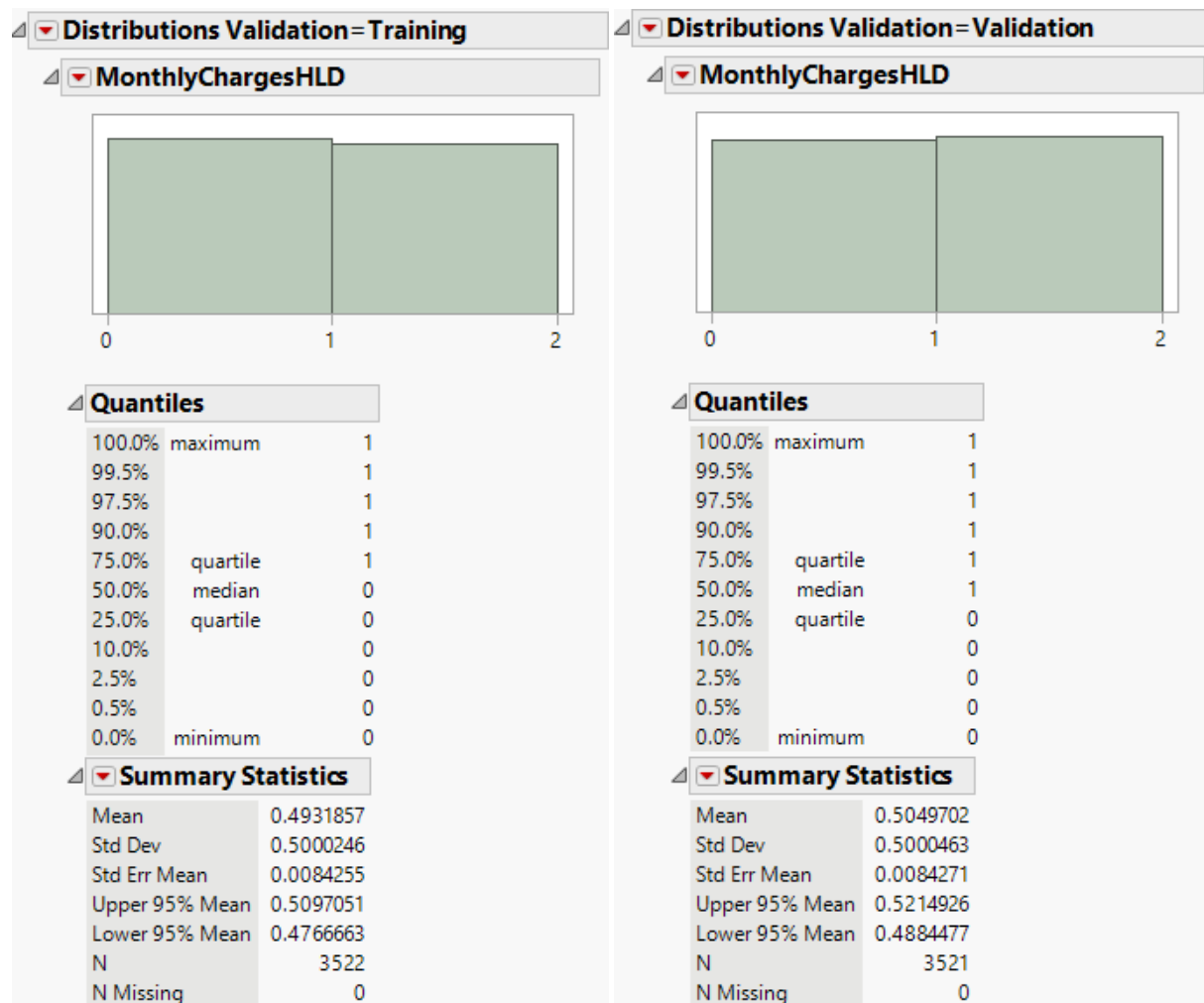
In this figure, we see the distributions of the dummy variable for Contract in both the training and validation sets. We see here that the distributions are virtually identical, with nearly the same counts (3522 in the training set and 3521 in the validation set) and nearly identical key statistics.

Figure 12 – Paperless Billing Dummy Variable Training-Validation Split



In this figure, we see the distributions of the dummy variable for Paperless Billing in both the training and validation sets. We see here that the distributions are virtually identical, with nearly the same counts (3522 in the training set and 3521 in the validation set) and nearly identical key statistics.

Figure 13 – Monthly Charges Dummy Variable Training-Validation Split

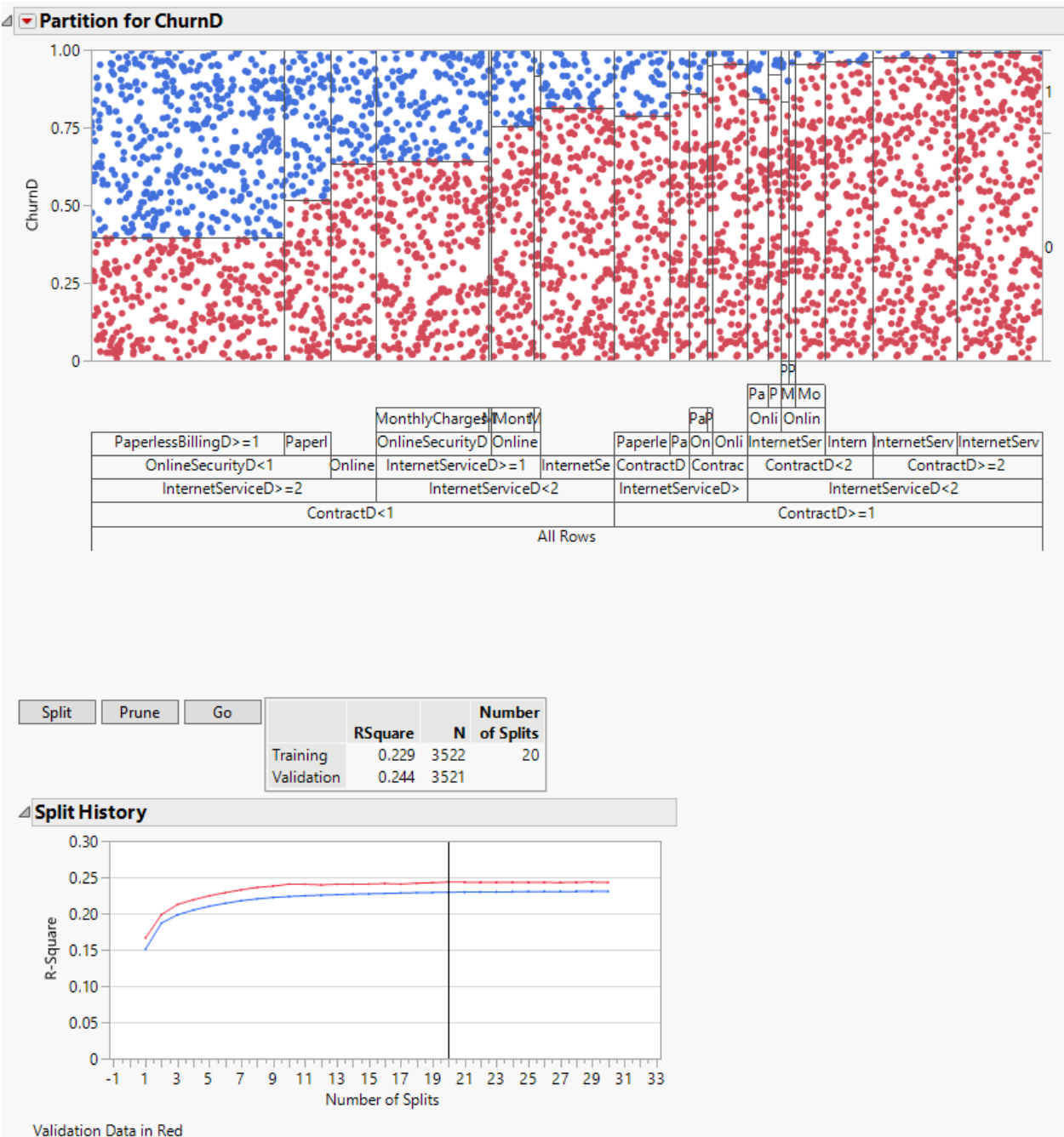


In this figure, we see the distributions of the dummy variable for Monthly Charges in both the training and validation sets. We see here that the distributions are virtually identical, with nearly the same counts (3522 in the training set and 3521 in the validation set) and nearly identical key statistics.

Modeling

We use decision tree modeling to analyze our data. This method breaks the data into smaller, homogeneous groups based on the target variable, which is customer churn in our case. For our model, we set our customer churn as our y-variable, and our five predictor variables as our x-variables. The validation column was set to the training-validation split data that was generated earlier.

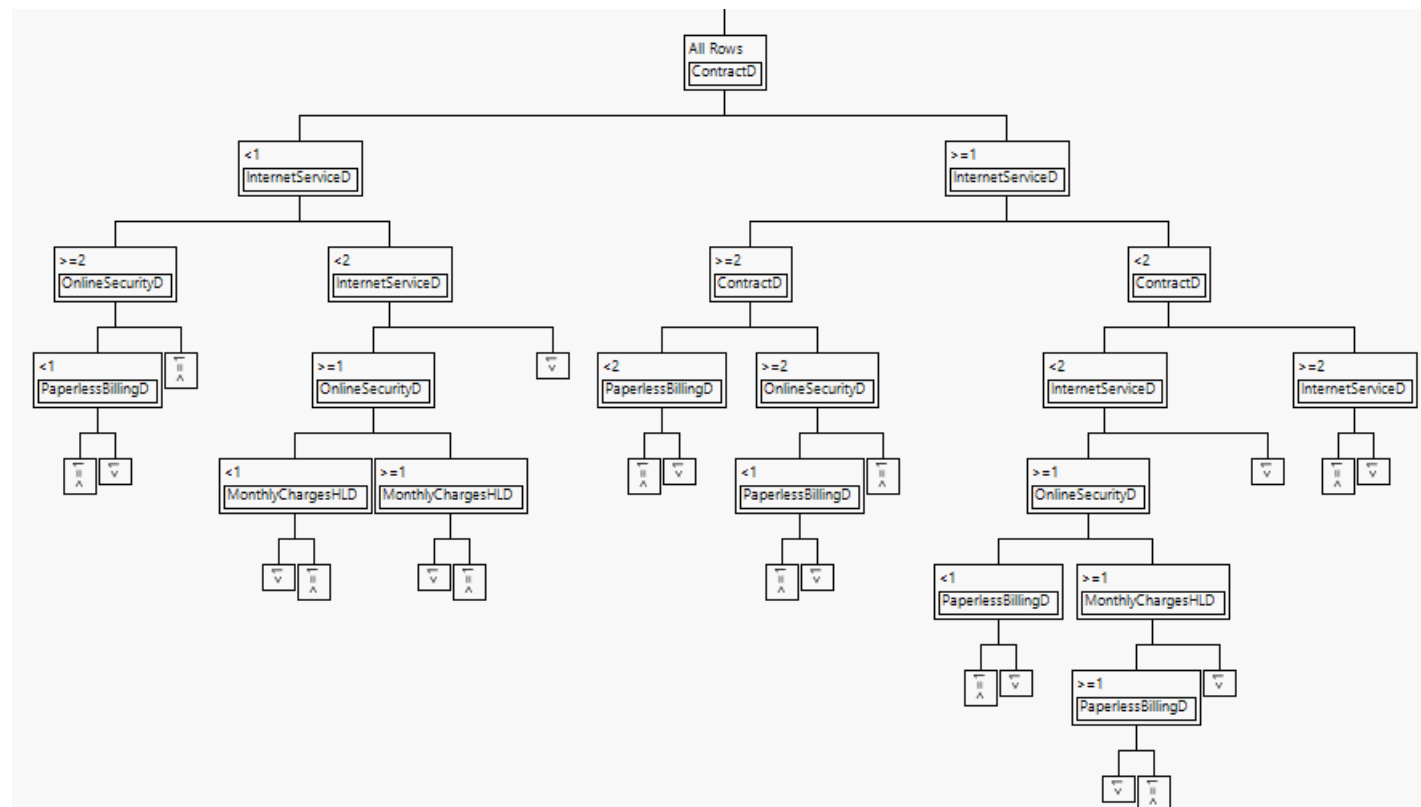
Figure 14 – Partition for Churn Data



In this figure, we see details about our decision tree model. We see the partition plot at the top, with horizontal lines indicating the overall proportion of the first plotted response level for each node of our model. The diagram below the plot shows what if-else statements each split consists of, and the if-else statements correspond to the vertical lines in the partition plot. The diagram is condensed, making it difficult to see each if-else statement,

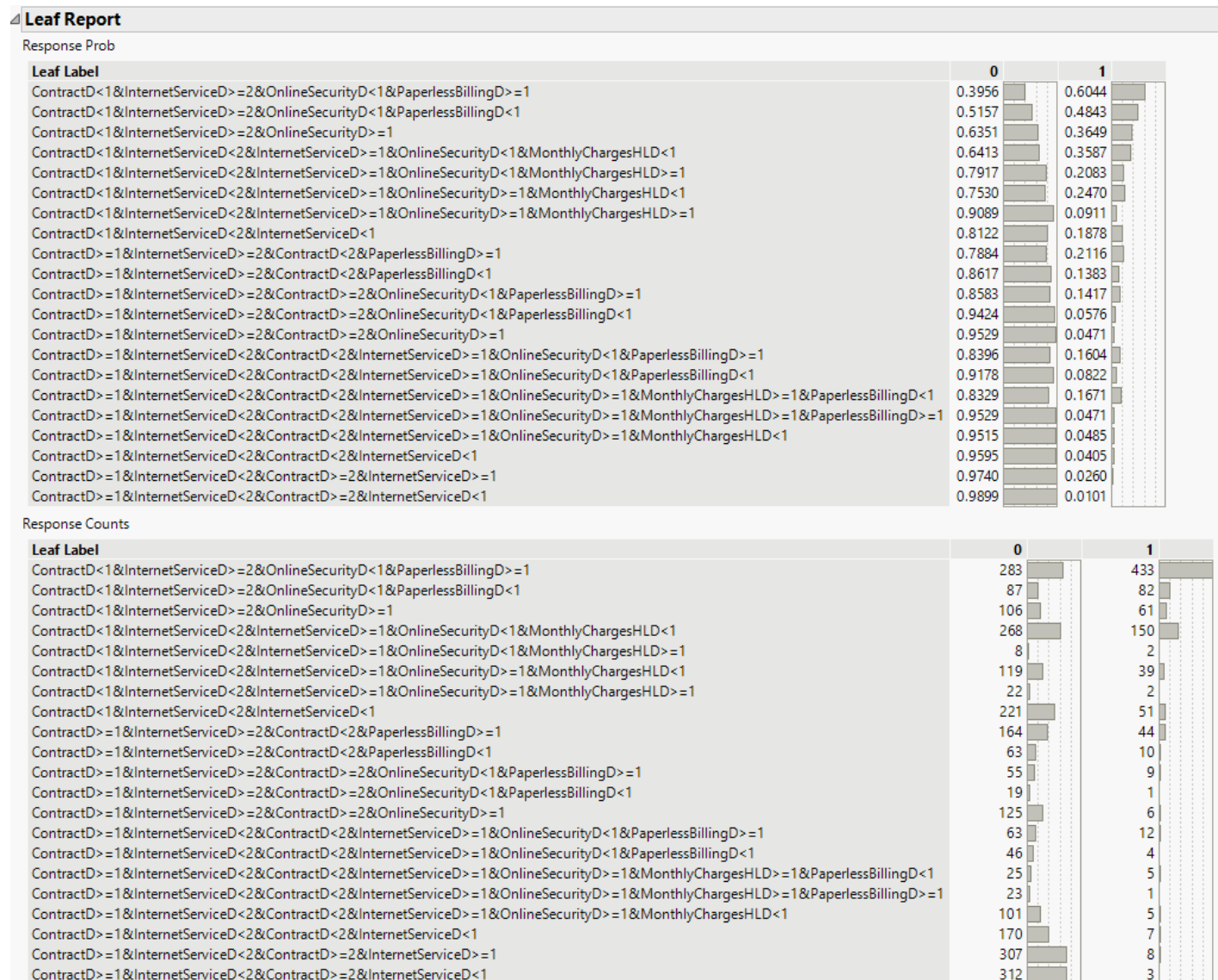
and so we will show a better visual representation of the model shortly. The split history graph shows that the R-squared values of our model for both the training and validation sets got to a value of 0.229 and 0.244 respectively, leveled out around 10 splits, and did not improve much afterwards, suggesting that it was appropriate to not increase the complexity of the model unnecessarily by going past 20 splits.

Figure 15 – Condensed Decision Tree



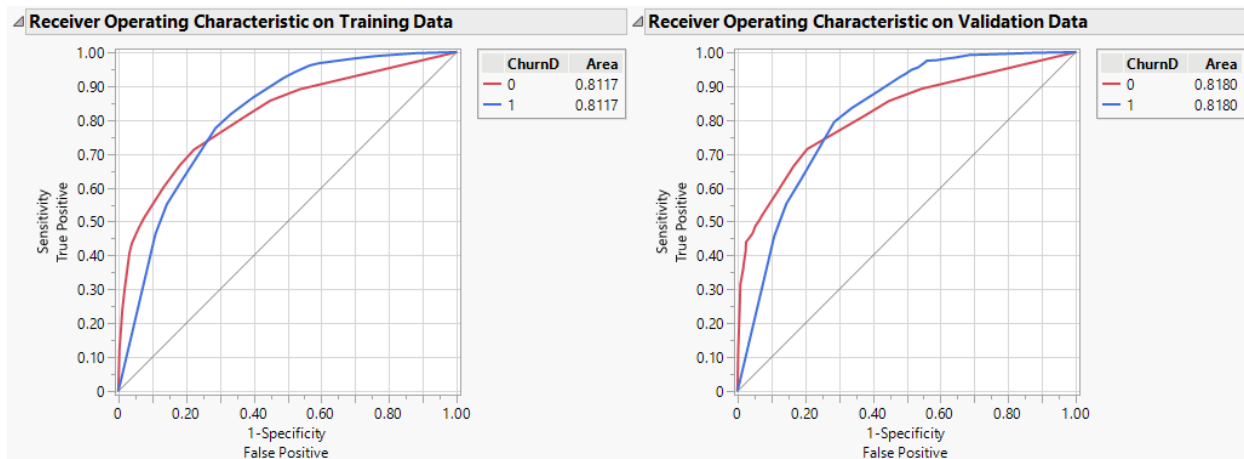
This figure shows our decision tree, in a somewhat condensed view so that it can be viewed in its entirety. We observe how splits were made using each of our five predictor variables to reach a conclusion for which combinations of splits were likely to lead to customer churn or retention.

Figure 16 – Leaf Report



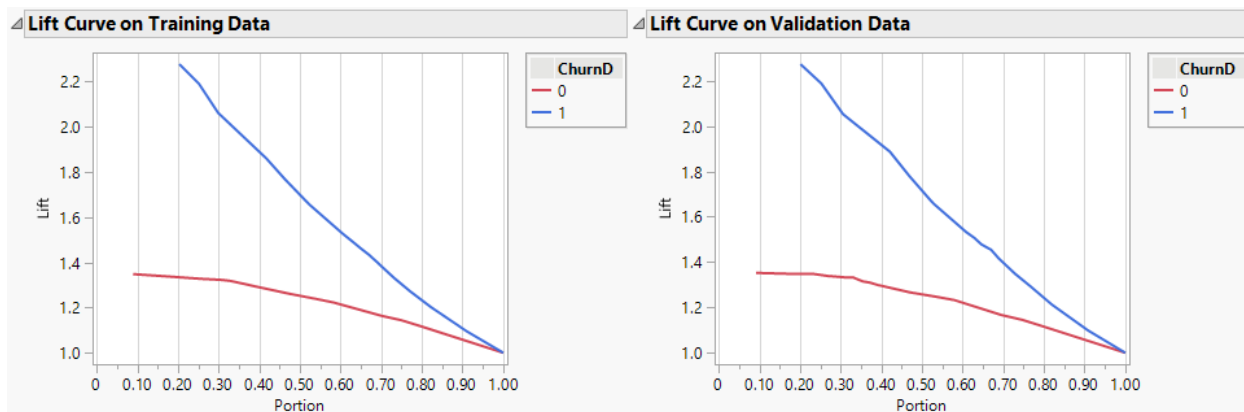
In this figure, we see the leaf report, which gives us greater insight into the decision tree shown in Figure 15. We see how each leaf of the decision tree, consisting of a variety of predictor variables, is associated with our target variable of churn. Each leaf, along with the values of predictor variables that make it up, is useful because we get to see which combinations of predictor variables and their values are associated with more or less churn.

Figure 17 – ROC Curve



This figure shows us the ROC (Receiver Operating Characteristic) curves for both the training and the validation data. The area under a ROC curve measures the degree of the goodness of fit for the decision tree to the measured data, with a value of 1 indicating that the model is a perfect fit while a value of 0.5 indicates that the model is no better than a random performance. Because the area under the ROC curve is slightly greater than 0.8 for both values of the churn variable in both sets, we conclude that the model is an excellent fit for our data.

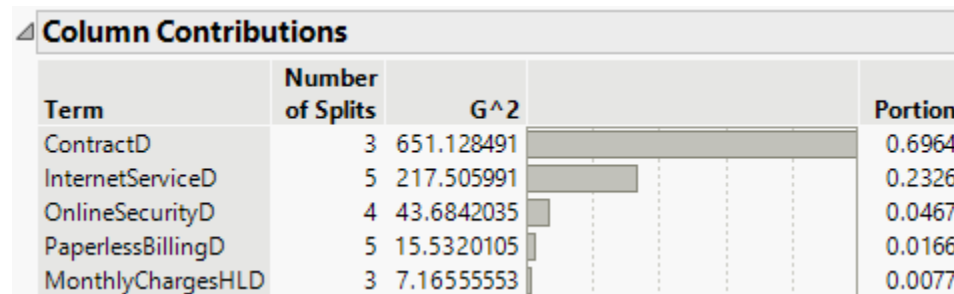
Figure 18 – Lift Curve



This figure shows us the lift curves for both the training and the validation data. Lift shows how much better the model is at predicting events compared to a random performance within certain percentiles. For example, a lift of roughly 2.2 at the 20th percentile means that the model is 120% better than random at predicting the associated event within the

20% of the observations with the highest predicted values. With a lift of 2.2 at the 20th percentile, using the model to choose the 20% of the observations with the highest predicted values will lead to having 2.2 times more responses where the ChurnD value is 1 (or the Churn value is Yes) compared to those 20% of observations being chosen randomly. The relatively high lifts in our figure indicate that the model seems to perform significantly better than a random performance.

Figure 18 – Column Contributions



This figure shows how much each column contributed to our model. We see that all five of our selected predictor variables were involved in the model. We also see to what degree each predictor was involved in the model: in order of most involved to least, we have Contract, Internet Service, Online Security, Paperless Billing, and Monthly Charges, with Contract and Internet Service making up a combined total of 92.90% of the model. This is significant, because if we wish to decrease the complexity of our model, we would start by removing the predictors that contributed least to the model.

Evaluation

We observe from our modeling process that the five predictor variables were significant to the model to varying degrees. Before assessing each of the five predictors, we will first assess the model's accuracy in predicting customer churn, and consider how our findings affect our strategies for customer retention.

Figure 19 – Confusion Matrix

Confusion Matrix

Training

Actual ChurnD	Predicted Count	
	0	1
0	2304	283
1	502	433

Validation

Actual ChurnD	Predicted Count	
	0	1
0	2306	281
1	507	427

Actual ChurnD	Predicted Rate	
	0	1
0	0.891	0.109
1	0.537	0.463

Actual ChurnD	Predicted Rate	
	0	1
0	0.891	0.109
1	0.543	0.457

This figure displays our confusion matrix, which shows a cross-tab of the observed values of churn versus the predicted values of churn, helping us to understand our model's predictive accuracy. We would want to calculate the **misclassification rate** of our model, or the rate at which our model incorrectly predicts the observed values. In our training set, the misclassification rate is $(283+502)/(2304+283+502+433) = 0.223$, or 22.3%. The misclassification rate for the validation set is similar: $(281+507)/(2306+281+507+427) = 0.224$, or 22.4%. Generally, a misclassification rate that is above 20% is considered to be high, since over a fifth of the observations are being predicted incorrectly. This gives us reason to have a degree of doubt regarding the accuracy of our predictive model.

The **sensitivity** of our model for the training set is $2304/(2304+283) = 0.891$ or 89.1%, and it is $2306/(2306+281) = 0.891$ or 89.1% for the validation set. This is a high sensitivity value, indicating that our model has relatively good performance when it comes to accurately predicting when churn did not occur. Our model would thus be useful in developing customer retention strategies in the aspect of identifying which predictor variables are associated with a customer not churning.

The **specificity** of our model for the training set is $433/(502+433) = 0.463$ or 46.3%, and it is $427/(507+427) = 0.457$ or 45.7% for the validation set. This is not a high specificity value, indicating that our model has relatively poor performance when it comes to accurately predicting when churn occurs. Our model would thus not be very useful in developing customer retention strategies in the aspect of identifying which predictor variables are associated with a customer churning.

Deployment

In conclusion, we would like to suggest that this model be used when seeking to identify which service and account features are most associated with customer churn and in the development of customer retention strategies, with certain limitations. Firstly, we observe from Figure 17 that the area under our ROC curves exceeded a value of 0.8, indicating that there is an excellent goodness of fit for our decision tree to the data, and therefore that our model is likely to be useful for predictive means. Secondly, we observe relatively high lift values in Figure 18, giving us cause to believe that our model is likely to be significantly better at predicting customer churn than a random performance. Thirdly, we note from Figure 19 that our misclassification rate of roughly 22.3% is relatively high, giving us cause to doubt the overall accuracy of our model. However, we also see from Figure 19 that our model has a high sensitivity, indicating that our model could be useful when it comes to identifying a customer that will not churn. The corresponding specificity in Figure 19 is low, indicating that our model is inaccurate when it comes to predicting customer churn. Overall, we believe that our model is an effective tool of prediction regarding customer churn depending on the context with notable flaws in its accuracy.

One suggestion for using this model comes from Figure 18. We see in this figure that the type of contract and the type of internet service were by far the most significant components of our model, when compared to the the other three predictors. We would recommend that these two variables be valued above the other three when seeking to predict customer churn, and if the company were to generate additional models with reduced complexity, that the company favor type of contract and type of internet service as predictor variables and exclude the other three predictors before these two.

A second suggestion would be to rerun the model with different predictor variables. If the company possesses the resources to do so, it could be beneficial to include a far larger variety of predictor variables, and then to optimize the model through a step-by-step splitting process. Manually developing the decision tree with a larger quantity of potential predictors could lead to a model with a misclassification rate low enough that the model's inaccuracy would not have to be a consideration in its deployment.

Thirdly, we would suggest that the current model's usage be limited to predicting customer retention and not customer churn. Figure 19 makes it clear from the high sensitivity rate and the low specificity rate that our model may be accurate when it comes to predicting a customer staying with the company, but that the model is inaccurate when it comes to predicting a customer churning. When developing customer retention strategies, the current model could be used to identify service and account features that are associated with a customer staying with the company, and therefore could be useful in identifying which features to promote and give advertisements about to potentially increase the percentage of customers that are retained. However, when seeking to identify which features are associated with customer churn, the current model has been shown to be inaccurate, and therefore it would be better to use a different type of predictive model or to modify the current one using the methods listed in our previous suggestion before attempting to identify which features are associated with customer churn.

Lastly, a practical application of using the model for customer retention could involve assessing the leaf report in Figure 16, which seems to indicate that having a one-to-two-year contract is associated with greater customer retention, as opposed to having a month-to-month contract. Based off of this observation, customer retention strategy could involve giving promotions and offering benefits regarding having a one-to-two-year contract. Different types of statistical analysis, such as logistic regression, could be used to verify that contract type and other variables are significant predictors of customer churn. We hope that this analytical report will be beneficial to the development of the company's customer retention strategies.